

Homework #5 (100 points)

207.547 Seminar in Experimental Psychology: Computational Modeling
Spring 2026

Due: Tuesday, May 19th, at 11:59 pm

If you finish the assignment, compress the whole directory into a single zip file that includes copies of your modified R codes and a single PDF file for questions below. Make sure your codes should run without an error when the TA or the instructor tests your codes. And make sure to check and report the convergence of model parameters for all questions.

1. (50 pts). Learn how to use the hBayesDM package and get some ideas about hierarchical Bayesian analysis.

Note: To learn how to use hBayesDM, checkout its online tutorial (https://ccs-lab.github.io/hBayesDM/articles/getting_started.html) and also this paper (https://www.mitpressjournals.org/doi/full/10.1162/CPSY_a_00002). Use four chains, 4000 MCMC samples, and 1000 warm-up samples. Note that installation (of rstan) is often the most challenging part. Check its manual (<https://github.com/stan-dev/rstan/wiki/RStan-Getting-Started>) carefully if you experience difficulty installing rstan or hBayesDM.

- (1) (30 pts) In HW2, you estimated the parameters of the Prospect model for the risk aversion task with individual maximum likelihood estimation (MLE).

Use the *ra_prospect* model in hBayesDM to estimate posterior distributions of the five subjects you used for HW#2, Question 2. Show the posterior distributions of the five subjects and compare MLE estimates (from HW2) and posterior means by plotting them in x- and y-axes.

- (2) (10 pts) Visually compare posterior means of the five subjects from MLE (from HW2) with those with hBayesDM by plotting them in x- and y-axes.
- (3) (10 pts) Discuss why/if two approaches led to (slightly) different results.

2. (20 pts) In HW#2, Question 3, you were asked to estimate model parameters of *ra_prospect* (model 1) and *ra_noLA* (model 2) models using MLE and computed BIC values.

- a) (5 pts) Approximate the Bayes factor of the two models (BF_{12}) from BIC values.
- b) (2 pts) According to Jefferys' evidence categories, how much evidence do we have for model 1 compared to model 2?
- c) (5 pts) If $p(M_1) = 0.6$ and $p(M_2) = 0.4$, what's posterior odds, $\frac{p(M_1|D)}{p(M_2|D)}$?
- d) (3 pts) What's the posterior model probability of model 1, $p(M_1|D)$?
- e) (5 pts) Approximate the Bayes Factor of seven memory retention models (HW2, Question 1) from their BIC values

3. (30pts). Stan_stan_codes_2026.zip file is shared through eTL.
Modify *ra_prospect_stan_singleSubj.R* and

`ra_prospect_stan_singleSubj.stan` codes so that

- (1) **(20 pts)** Your code will calculate posterior distributions of all 5 subjects (Hint: use a *for* loop within Stan)
- (2) **(10 pts)** Calculate posterior means of 5 subjects (Hint: `mean(...)`)